

The schematic diagram illustrates the MotoRecoPico board, which is a custom PCB designed for a Raspberry Pi Pico 2. The board features a central MCU (Raspberry Pi Pico 2) and several peripheral components:

- Power Management:** A 12V input is connected to a high-side switch (Q2, ZXMP6A17E6Q) and a 5V buck converter (U4, AP63205WU-7). The buck converter outputs +5V, which is decoupled with capacitors (C15, C13). A 3V3 supply is also provided for the MCU and other components.
- CAN Bus:** The MCP25625 CAN transceiver (U2) is connected to the CANH and CANL lines. It is powered by +5V and +3V3, with decoupling capacitors (C3, C4, C5, C6). The MCU's CAN pins (GPIO10, GPIO11, GPIO12, GPIO13) are connected to the transceiver.
- MicroSD:** A microSD card (DM3AT) is connected to the MCU's SPI pins (GPIO16, GPIO17, GPIO18, GPIO19, GPIO20, GPIO21, GPIO22). The card is powered by +3V3 and has a pull-up resistor (R10) on the SD_CS pin.
- GPS:** The GT-502MGG-N GPS module is connected to the MCU's UART pins (GPS_TX, GPS_RX, GPS_PPS). It is powered by +3V3.
- LED:** A status LED (GREEN) is connected to the MCU's LED_STATUS pin (GPIO14) and a 1k resistor (R14).
- Other Components:** The board includes a 10k pull-up resistor (R1) on the IG_RAW line, a 100nF capacitor (C14) on the EN pin of the buck converter, and various other passive components for decoupling and timing.

The schematic is a detailed representation of the board's layout, showing the physical placement of components and their electrical connections. It is a KiCad project, as indicated by the file name "MotoRecoPico.kicad_sch".

Sheet: /		
File: MotoRecoPico.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 1/1